



NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCE

CS 6120 — Assignment 1

Due: September 21, 2026 (100 points)

YOUR NAME + E-MAIL

In this assignment, we will develop a *word completion* autocomplete system that assists creative writing by suggesting words, phrases, or even entire sentences based on the style, genre, or theme of Shakespeare. To do so, we'll need the complete works of Shakespeare from Project Gutenberg, which is available at [our website with slight modifications](#). When you've finished with this assignment, turn in your code and its artifacts on Gradescope. Your checklist:

- Python file named `autocomplete.py`, with the following functions
 - ☐ Your solution in Q1: `read_vocabulary`
 - ☐ Your final solution in Q2: `autocomplete_word`

Question 1: Setup and Preprocessing

In this question, we will do all the text processing that will enable us to work on the common problem of auto-completing words.

Download the data. If you're working in Google Colab, you can do so using the `!` magick - `!<command>`, which allows you to use Linux commands in the cell. In this case, the linux command is `wget --no-check-certificate`. In an execution cell, the following command will download the data to a local folder in the workspace:

```
!wget --no-check-certificate https://course.ccs.neu.edu/cs6120s26/data/shakespeare/shakespeare-edit.txt
```

Look for the download in your workspace from Project Gutenberg that is called `shakespeare-edit.txt`, the complete works, including sonnets, of William Shakespeare.

Write a function that:

- reads in the data through UTF-8
- removes all punctuation
- lower-cases every letter

- splits by space to create all words
- removes numerical digits
- returns a list of unique words ordered by their frequency (most frequent to least)

The function signature should look like this:

```
def read_vocabulary(filename):
    """
    Reads in a given file specified by "filename" and processes it
    by removing punctuation, forcing lowercase, splits into
    individual words, and removes the numbers that might appear in
    the text.

    Args:
        filename: the name of the file to be processed

    Returns:
        A list of words in the order in which they appeared in the
        text.
    """
    % <YOUR-CODE-HERE>
    return []
```

The result of the above should produce a list with words in the order by frequency.

```
['the',
 'and',
 'i',
 '...']
```

Question 2: Autocomplete

In this question, we will write the function `autocomplete`, which takes a string prefix and the `model_or_data_structure`. To set up any data structures (to make the algorithm efficient), we have provided the function `process_data` that takes in the word frequencies created in the last function. Because this course is about its applications in practice, can you make your solution computationally fast? The function signature is as follows:

```
def process_data(word_list):
    """
    Builds any data structure or model from a list of words.

    Args:
        words: A list of words.

    Returns:
        Any_data_structure_you_like
    """
    <YOUR-CODE-HERE>
    return model_or_data_structure
```

```
def autocomplete_word(prefix, model_or_data_structure):
    """
    Returns a list of words starting with the given prefix. This list
    is sorted in order of the frequency (probability) of the words.

    Args:
        prefix: The prefix to search for.
        model_or_data_structure: model or data, considering the frequency
                                of word occurrence.

    Returns:
        A list of ten most-common words starting with the prefix.
    """

    return []
```

Testing Out Your Code

Test your functions out in real-time. You can do that in a number of ways. For example, if you are in Google Colab, the following code will create a text box that updates everytime you enter a character:

```
import ipywidgets as widgets
from IPython.display import display

def on_value_change(change):
    prefix = change['new']
    suggestions = autocomplete_word(prefix, model_or_data_structure)
    if suggestions:
        with output:
            output.clear_output()
            print("Suggestions:")
            for word in suggestions:
                print(word)
    else:
        with output:
            output.clear_output()
            print("No suggestions found.")

text = widgets.Text()
output = widgets.Output()

display(text, output)
text.observe(on_value_change, names='value')
```

Or, if you would like to get a head start on making an app, the following code will create a real-time [streamlit](#) app:

```
import streamlit as st
from st_keyup import st_keyup
```

<YOUR-CODE-HERE>

```
def main():
    # Notice value updates after every key press
    st.title("Autocomplete App")
    query = st_keyup("Enter a value", debounce=500, key="0")
    suggestions = autocomplete(query, model_or_data_structure)
    st.write(suggestions)
```

You would need the streamlit and st_keyup libraries. You can install with pip.

Submission Instructions

Upload your artifact (assignment1.py to Gradescope.

Code All your code saved in a Python file called assignment1.py. We will be expecting multiple different functions in assignment1.py, including:

- Q1 - read_vocabulary
- Q2 - autocomplete_word
- Any code used to test your algorithms
- Include comments you have for how your algorithm works, and some test results